

Project Initialization and Planning Phase

Date	27 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	3 Marks

Project Proposal

Project Overview	
Objective	To develop a machine learning-based system that accurately detects fraudulent online payment transactions in real-time, integrated with a Flask web application to enable users to input transaction details and receive instant fraud predictions, thereby enhancing financial security and reducing losses.
Scope	Develop and deploy a ML algorithm for fraud detection using a balanced dataset of transaction records. Implement a Flask web application with user authentication (registration, login via email, Gmail, and Facebook) and transaction input/prediction functionality.
Problem Statement	
Description	Online payment fraud poses a significant challenge for businesses and financial institutions, as fraudulent transactions are rare but costly, often going undetected until substantial losses occur. Small business owners and financial managers lack accessible, real-time tools to identify suspicious transactions efficiently, relying on manual reviews or systems that generate excessive false positives, which disrupt legitimate transactions and erode customer trust.
Impact	Addressing this problem will:

	● Reduce financial losses by enabling early detection of fraudulent transactions. ● Enhance customer trust and satisfaction through secure payment processing. ● Streamline operational efficiency by automating fraud detection processes. ● Support financial institutions in meeting regulatory compliance requirements by minimizing fraud risks.
Proposed Solution	
Approach	1. Data Preprocessing: Downsample non-fraudulent transactions to balance the dataset (1,288 non-fraudulent and 1,142 fraudulent records), encode categorical features (e.g., transaction type using LabelEncoder), and drop irrelevant features (nameOrig, nameDest). 2. Model Development: Train and evaluate multiple machine learning models (Random Forest, XGBoost, Decision Tree, Extra Trees, SVC) on the balanced dataset, selecting the Random Forest Classifier for deployment due to its high accuracy (97.33%) and robustness. 3. Testing and Validation: Assess model performance using accuracy, precision, recall, and F1-score, and validate the web application for usability and reliability. 4. Web Application Development: Build a Flask-based web application with HTML templates (home.html, predict.html, submit.html) for user registration, login, and transaction input, integrating the trained model for real-time predictions.
Key Features	Real-Time Fraud Prediction: Users can input transaction details (step, type, amount, oldbalanceOrg, newbalanceOrig, oldbalanceDest, newbalanceDest) to receive instant fraud predictions. User-Friendly Interface: Flask-based web application with simple easy to understand templates.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for model training and Flask server	Standard CPU (e.g., Intel i7)
Memory	RAM for processing	8 GB
Storage	Disk space for data, models, and logs	500 GB SSD
Software		
Frameworks	Python web frameworks	Flask
Libraries	Machine learning and data processing	scikit-learn, pandas, numpy, xgboost, pickle
Development Environment	IDE, version control	Jupyter notebook, Git
Data		
Data	Transaction dataset -1,048,575 rows, 10 columns.	Kaggle dataset